

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000812		
	Version:	8.0; Most-Recent; Effective; CURRENT		
	Status:	Effective	Purpose:	Uncontrolled Copy
Class:	Control del Diseño		Effective date:	10-oct-2013
Title:	SM DG57. Reader camera adjustment.			

DG-57 Service Manual

READER CAMERA ADJUSTMENT

Procedure description
This procedure describes how to adjust Reader camera with Matrox Meteor II and Matrox Morphis PCI Express.

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EQUIPMENT

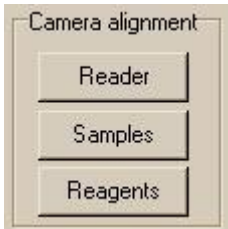
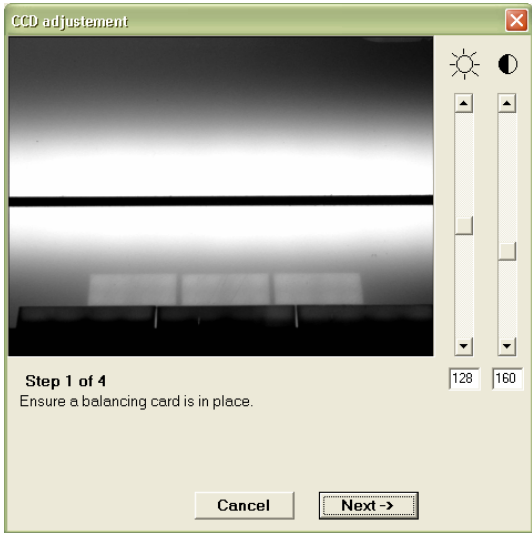
1. An instrument connected with a PC with the software installed.
2. Brand new Neutral (NaCl) card

SET UP

1. Turn on the instrument.
2. Start the ST software (Technical Service software).
3. Enter the right Technical Service password (d+g_diana).

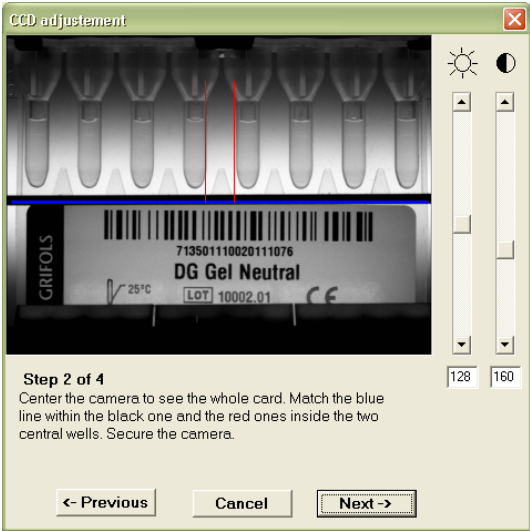
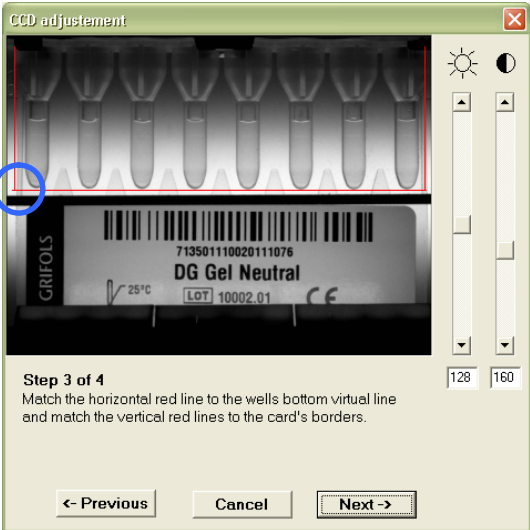
PROCEDURE

NOTE: Time to perform procedure between steps 1 and 20 must not exceed 2minutes.

Process	Expected Results
Adjusting Reader camera position 1. In <i>Settings</i> Tab, click on <i>Reader</i> button in <i>Camera alignment</i> section.  Picture 1 2. Place a brand new Neutral (NaCl) card in balancing card.  Picture 2 3. Click on <i>Next</i>.	New window appears (Picture 2). --- New window appears (Picture 3). The gripper takes the card in balancing card position to the reading position.

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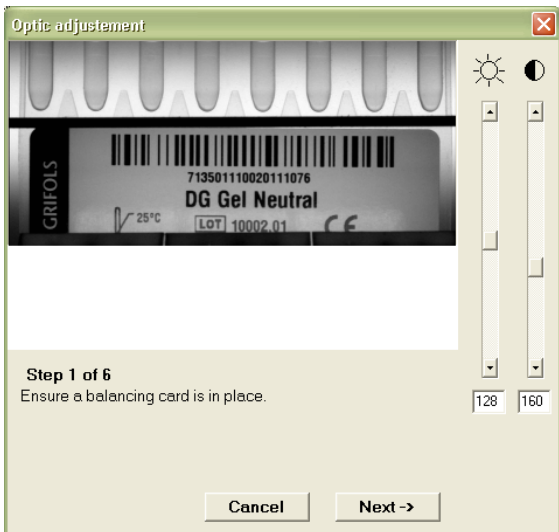
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Process	Expected Results
4. Move the camera right/left to center the red lines between central wells. Move the camera up/down to match the blue line with the black one.  Picture 3 5. Click on <i>Next</i>. 6. Position the U-shaped red lines so the bottom line touches the bottom of all wells (just touching, not crossing) and the vertical lines match the edges of the card. That is done by locating one of the vertexes of the “U” in the right position (see blue circle in the image). The other vertex will position automatically, symmetrically to the centre of the image.  Picture 4 7. Click on <i>Next</i>. A new window appears. Click on <i>Finish</i>. Adjusting Reader camera optics 8. Place a brand new Neutral (NaCl) card in balancing card position. 9. Disable temperature control: access I/O tab and click on <i>Off</i> checkbox in <i>Heating control</i> section.	--- New window appears (Picture 4). --- --- --- ---

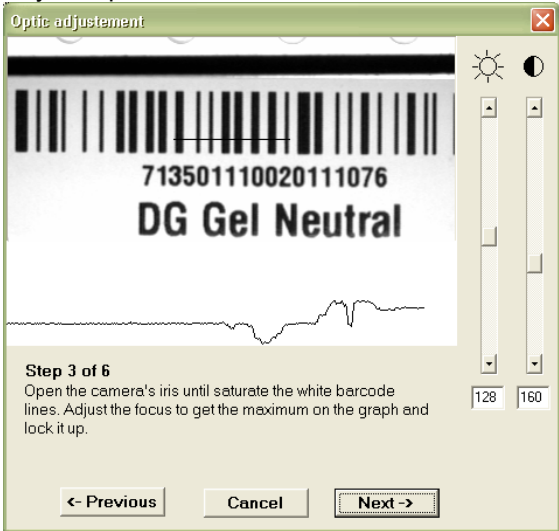
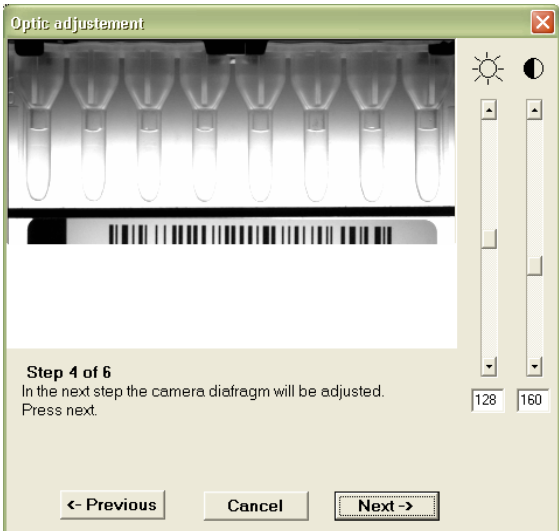
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Process	Expected Results
10. In <i>Settings</i> tab, click on <i>Reader</i> button in <i>Camera optics</i> section.  Picture 7 11. Follow the instructions on the screen. Place a card in position 1 of Balancing card rack.  Picture 8 12. Click on <i>Next</i>. 13. Centre the blue square on the barcode. The blue cross has to be in the center of the barcode.  Picture 9	New window appears (Picture 8). --- New window appears (Picture 9). ---

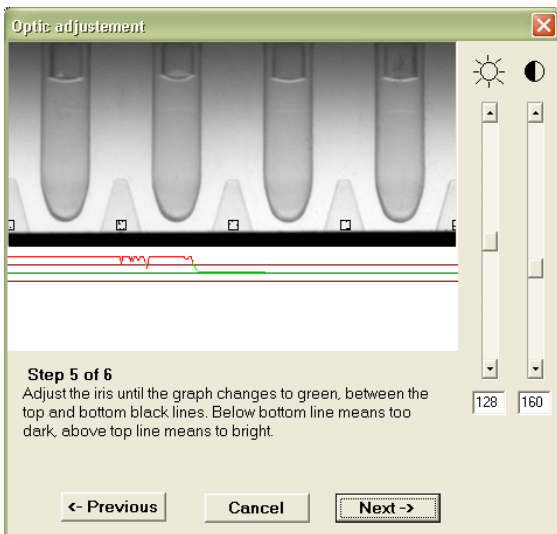
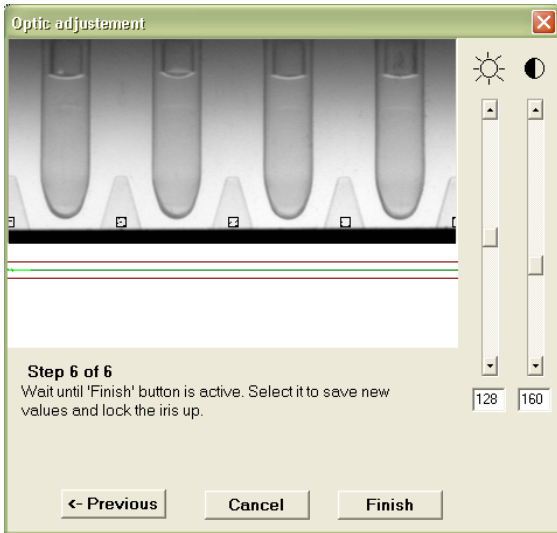
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Process	Expected Results
14. Click on <i>Next</i>. 15. Move the camera focus until the graph reaches its maximum position. Depending on many factors the graph may get higher or lower, but it is possible to find only one peak.  Picture 10 16. Click on <i>Next</i>. 17. When ready click on <i>Next</i> to begin camera iris adjustment.  Picture 11 18. Move the camera iris until the graph turns to green. Ensure the graph is on the central line or above.	New window appears (Picture 10). --- New window appears (Picture 11). New window appears (Picture 12). ---

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Process	Expected Results												
 Picture 12 19. Click on <i>Next</i>. 20. Check that brightness value is between: <table><tr><th>Matrox Board</th><th>Lower limit</th><th>Upper limit</th><th>Contrast</th></tr><tr><td>Meteor II</td><td>123</td><td>128</td><td>160</td></tr><tr><td>Morphis</td><td>112</td><td>114</td><td>166</td></tr></table> When <i>Finish</i> button becomes available, click on it to end procedure and save new settings.  Picture 13 Fine adjustment of Reader camera optics	Matrox Board	Lower limit	Upper limit	Contrast	Meteor II	123	128	160	Morphis	112	114	166	New window appears (Picture 12). ---
Matrox Board	Lower limit	Upper limit	Contrast										
Meteor II	123	128	160										
Morphis	112	114	166										

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Process	Expected Results												
21. Run previous procedure at least 16 minutes after step 20. During this time, Reader camera illumination leds must be off.	---												
22. In <i>Settings</i> tab, click on <i>Reader</i> button in <i>Camera optics</i> section.	---												
23. Click on <i>Next</i> button 5 times until Picture 13 appears.	---												
24. Wait until Finish button is available. Check that brightness value is:	---												
<table><tr><th>Matrox Board</th><th>Lower limit</th><th>Upper limit</th><th>Contrast</th></tr><tr><td>Meteor II</td><td>100</td><td>128</td><td>160</td></tr><tr><td>Morphis</td><td>103</td><td>114</td><td>166</td></tr></table>		Matrox Board	Lower limit	Upper limit	Contrast	Meteor II	100	128	160	Morphis	103	114	166
Matrox Board	Lower limit	Upper limit	Contrast										
Meteor II	100	128	160										
Morphis	103	114	166										
if it is inside this margin push <i>Finish</i> button, if not repeat optic adjustments from step 9.													
Checking Reader camera adjustment													
25. Run a real batch.	---												
26. Check that all card barcodes are read correctly	---												
27. Check at the end of the batch that the reading given by the software is coincident with a manual reading.	---												

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DOCUMENT CHANGE HISTORY

Document change history			
Version	Date	Change	Rationale
4.0	2005/05/06	Document REGD-0000812_v4 replaces document 57040V0M100.	Uploaded document in GxPharma.
5.0	2010/09/21	Updated all the pictures of the document.	The document is updated due to the implementation of new functionalities.
5.0	2010/09/21	Document REGD-0000812_v5 replaces document REGD-0000812_v4.	Document version changed.
6.0	2011/06/28	Updated the Document Change History: -Date format changed to YYYY/MM/DD. -Dates changed to document version effective date.	Adoption of ISO 8601 date format to avoid ambiguity.
6.0	2011/06/28	Added previous version 'data', 'change' and 'rationale' inside document change history section.	Update change history for previous document history modification.
6.0	2011/06/28	Document REGD-0000812_v6 replaces document REGD-0000812_v5.	Document version changed.
6.0	2011/06/28	Procedure updated.	Step 21 has been modified.
6.0	2011/06/28	Document title.	Commercial reference substituted by project reference.
6.0	2011/06/28	Document header.	Increase information to easy document search.
7.0	2013/03/25	Document title.	This document now only applies to DG-57
7.0	2013/03/25	Matrox Morphis brightness range.	Brightness range has been optimized for Matrox Morphis board.
8.0	See document header.	-Maximum time to perform procedure has been established. -Waiting time before 'fine adjustment' has been updated. -brightness contrast is indicated.	Ensure correct adjustment.

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